

A Counterexample to Conjecture 00000000046: the Prime-Sum Graphs Stay at Cut Distance $\geq 1/8$ from $W \equiv 1/2$

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Abstract

Conjecture 00000000046 asserts that the prime-sum graphs G_n converge, in the sense of graph limits (cut distance), to the constant graphon $W \equiv 1/2$. They do not: $\delta_{\square}(W_{G_n}, 1/2) \geq 1/8$ for every $n \geq 1$. The odd vertices of G_n are pairwise non-adjacent and carry at least half of $[0, 1]$, so the step graphon vanishes on a box of measure at least $1/4$. The refutation is formalized in Lean 4 against Mathlib, including the step graphon, the cut norm, the cut distance and the convergence statement itself.

1 The conjecture as stated

The original text of conjecture 00000000046 reads:

Conjecture: G_n converges, in the sense of graph limits (cut distance), to the constant graphon $W \equiv 1/2$.

2 Reading of the statement

The graphs. G_n is the prime-sum graph of the preceding conjecture 00000000045: its vertices are $1, \dots, n$, and $i \sim j$ if and only if $i \neq j$ and $i + j$ is prime.

Graph limits. We use the standard definitions of the theory of dense graph limits [1], on $[0, 1]$ with Lebesgue measure λ .

Definition 1. 1. A *graphon* is a measurable symmetric function $W : [0, 1]^2 \rightarrow [0, 1]$.

2. The *step graphon* of a graph G on $\{1, \dots, n\}$ is $W_G(x, y) = 1$ if the vertices carrying x and y are adjacent and 0 otherwise. Here vertex i carries the i -th interval of the partition $[0, \frac{1}{n}], (\frac{1}{n}, \frac{2}{n}], \dots, (\frac{n-1}{n}, 1]$, that is, x is carried by vertex $\max(1, \lceil nx \rceil)$.

3. The *cut norm* of a bounded measurable $U : [0, 1]^2 \rightarrow \mathbb{R}$ is

$$\|U\|_{\square} = \sup_{S, T} \left| \int_{S \times T} U \, d\lambda^2 \right|,$$

the supremum over measurable $S, T \subseteq [0, 1]$.

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4. For $\varphi: [0, 1] \rightarrow [0, 1]$ put $W^\varphi(x, y) = W(\varphi x, \varphi y)$. The *cut distance* is

$$\delta_\square(U, W) = \inf_\varphi \|U - W^\varphi\|_\square,$$

the infimum over invertible measure-preserving maps $\varphi: [0, 1] \rightarrow [0, 1]$.

5. A graph sequence (G_n) *converges* to a graphon W if $\delta_\square(W_{G_n}, W) \rightarrow 0$.

So the conjecture asserts that $\delta_\square(W_{G_n}, \frac{1}{2}) \rightarrow 0$ as $n \rightarrow \infty$, where $\frac{1}{2}$ denotes the constant graphon. Other common forms of the cut distance (relabelling either argument, or taking the infimum over couplings) give the same conclusion; see Remark 6.

3 Result

Theorem 2. *For every $n \geq 1$, $\delta_\square(W_{G_n}, \frac{1}{2}) \geq \frac{1}{8}$. In particular G_n does not converge to the constant graphon $\frac{1}{2}$, and Conjecture 00000000046 is false.*

4 Proof

Lemma 3. *For every bounded measurable U , $\delta_\square(U, \frac{1}{2}) = \|U - \frac{1}{2}\|_\square$.*

Proof. A constant kernel is unchanged by relabelling, so every term of the infimum equals $\|U - \frac{1}{2}\|_\square$; the infimum is over a nonempty set, since the identity is an invertible measure-preserving map. $\square$

Lemma 4. *The odd vertices of G_n are pairwise non-adjacent.*

Proof. If $i \neq j$ are both odd, then $i + j$ is even and at least $1 + 3 = 4$, so it is not prime. $\square$

Lemma 5. *Let $n \geq 1$ and let $S_n \subseteq [0, 1]$ be the union of the intervals carried by the odd vertices of G_n . Then $\lambda(S_n) = \lceil n/2 \rceil / n \geq \frac{1}{2}$.*

Proof. Each of the n intervals of the partition has length $1/n$, and $\{1, \dots, n\}$ contains $\lceil n/2 \rceil = \lfloor (n+1)/2 \rfloor$ odd numbers. $\square$

Proof of Theorem 2. By Lemma 4, $W_{G_n} = 0$ on $S_n \times S_n$, so $W_{G_n} - \frac{1}{2} = -\frac{1}{2}$ there. Taking $S = T = S_n$ in the definition of the cut norm and using Lemmas 3 and 5,

$$\delta_\square(W_{G_n}, \frac{1}{2}) = \|W_{G_n} - \frac{1}{2}\|_\square \geq \left| \int_{S_n \times S_n} (W_{G_n} - \frac{1}{2}) \right| = \frac{1}{2} \lambda(S_n)^2 \geq \frac{1}{8}.$$

Hence $\delta_\square(W_{G_n}, \frac{1}{2})$ does not tend to 0. $\square$

5 Formalization

The accompanying Lean 4 project (`lean/Submission/Basic.lean`) works on Mathlib's unit interval `unitInterval` with its Lebesgue measure `volume` (a probability measure), and on `unitInterval × unitInterval` with the product measure. It states the conjecture with the definitions above and proves its negation.

- `IsGraphon`, `cutNorm`, `relabel`, `cutDist` — items 1, 3 and 4 of the definition. The cut norm is a supremum over pairs of measurable sets and the cut distance an infimum over measurable bijections φ with measurable inverse (`MeasurableEquiv`) that preserve volume.

- `vtx n x` = $\max(1, \lceil nx \rceil)$, the vertex carrying x ; `PrimeSumAdj` — adjacency in G_n ; `W n` — the step graphon W_{G_n} ; `half` — the constant graphon. `isGraphon_W` and `isGraphon_half` check that both are graphons.
- `ConjectureHolds` — `cutDist (W n) half` tends to 0 as $n \rightarrow \infty$ (a `Tendsto` statement), item 5 of the definition.
- `volume_vtx_preimage`, `map_vtx` — every vertex carries an interval of measure $1/n$, so `vtx n` is uniformly distributed on $\{1, \dots, n\}$; `card_odd_Icc`, `half_le_volume_oddCells` — Lemma 5.
- `not_primeSumAdj_of_odd`, `W_eq_zero_of_mem` — Lemma 4 and its consequence for W_{G_n} ; `setIntegral_oddCells` — the value $-\frac{1}{2}\lambda(S_n)^2$ of the integral over $S_n \times S_n$.
- `abs_setIntegral_le_one` — every box integral of $W_{G_n} - \frac{1}{2}$ is at most 1 in absolute value, so the supremum in `cutNorm` is a genuine least upper bound; `eighth_le_cutNorm` — $\|W_{G_n} - \frac{1}{2}\|_{\square} \geq \frac{1}{8}$.
- `relabel_half`, `cutDist_half` — Lemma 3; `eighth_le_cutDist` — Theorem 2.
- `conjecture_00000000046_false` — $\neg \text{ConjectureHolds}$.

The toolchain is `leanprover/lean4:v4.33.1`, `Mathlib` is pinned to `v4.33.1`, and `lake build` succeeds. The project contains no `sorry`, no `admit` and no `native_decide`, and introduces no axioms:

```
#print axioms conjecture_00000000046_false
```

reports exactly `propext`, `Classical.choice` and `Quot.sound`.

6 Remarks

Remark 6 (other forms of the cut distance). The bound does not depend on the variant of $\delta_{\square}$. Relabelling the first argument instead, $\|U^{\varphi} - \frac{1}{2}\|_{\square} \geq |\int_{S' \times S'} (U^{\varphi} - \frac{1}{2})|$ with $S' = \varphi^{-1}(S_n)$, and the integral equals $\int_{S_n \times S_n} (U - \frac{1}{2})$ because φ preserves measure. For the definition by couplings μ of (λ, λ) , the box $(S_n \times [0, 1])^2$ gives the same integral, because the first marginal of μ is λ . In every case the distance is at least $\frac{1}{8}$.

Remark 7 (what G_n does converge to). The edge density of G_n tends to 0: every edge $\{i, j\}$ has $i + j$ a prime at most $2n$, and each such prime p accounts for fewer than $p/2 \leq n$ edges, so $e(G_n) \leq n \pi(2n) = O(n^2 / \log n)$. Since $\|W_{G_n}\|_{\square} \leq \int W_{G_n} = 2e(G_n)/n^2$, the sequence G_n converges to the zero graphon. This remark is not part of the Lean development.

Remark 8 (relation to an earlier submission). An earlier submission for this conjecture used the triangle-freeness of G_n and the counting lemma, but left the graph-limit part (graphons, cut norm, cut distance) unformalized, and it was withdrawn after review on 2026-10-01. The present argument is different and shorter. It uses a single box instead of homomorphism densities, and the Lean development states the conjecture with the graph-limit notions themselves and refutes it.

References

- [1] L. Lovász, *Large Networks and Graph Limits*, American Mathematical Society Colloquium Publications 60, AMS, 2012.